

AI Asset Vault – Project Overview & Progress Documentation

Project Title

AI Asset Vault – Intelligent Document Management System

Project Overview

AI Asset Vault is a full-stack web application designed to securely store, organize, and manage digital documents while enhancing the user experience with Artificial Intelligence. Instead of acting as a traditional file storage system, the application understands uploaded documents, generates AI-powered summaries, enables conversational interaction with document content, automatically generates relevant tags, and provides intelligent search capabilities.

The project is being developed using **FastAPI** for the backend, **PostgreSQL** as the database, **Google Gemini AI** for document intelligence, and **Next.js with Tailwind CSS** for the frontend.

Project Objectives

- Develop a secure document management platform.
 - Enable authenticated users to upload and manage their documents.
 - Automatically extract text from PDF files.
 - Generate AI-powered summaries for uploaded documents.
 - Allow users to ask questions about document content using AI.
 - Maintain chat history for every document.
 - Automatically generate intelligent tags for better organization.
 - Provide search functionality for quick document retrieval.
 - Build a modern, responsive, and user-friendly web interface.
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Technology Stack

Backend

- Python
- FastAPI
- SQLAlchemy
- PostgreSQL
- JWT Authentication
- Google Gemini API

Frontend

- Next.js
- React
- Tailwind CSS
- Axios

AI Features

- PDF Text Extraction
 - AI Document Summarization
 - AI Question Answering
 - AI Tag Generation
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Features Completed

User Authentication

- User Registration
- Secure Login
- JWT Token Authentication
- Protected API Endpoints

Asset Management

- Upload PDF documents
- Secure file storage
- Download documents
- Delete documents
- View uploaded assets

AI Features

- Automatic PDF text extraction
- AI-generated document summaries
- Chat with uploaded documents
- Chat history storage
- Automatic AI-generated tags

Search & Dashboard

- Search assets by filename
 - Dashboard statistics
 - Asset management APIs
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Database Structure

The project currently consists of the following primary database tables:

- Users
- Assets
- ChatHistory

These tables maintain user information, uploaded document metadata, AI summaries, extracted text, generated tags, and conversation history.

REST API Endpoints Implemented

Authentication

- Register User
- Login User
- Get Current User

Asset Management

- Upload Asset
- View My Assets
- Download Asset
- Delete Asset
- Search Assets
- Dashboard Statistics

AI Features

- Generate Document Summary
 - Chat with Document
 - Retrieve Chat History
 - Generate AI Tags
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Current Project Status

Backend

Completed

The backend implementation includes authentication, database integration, AI services, document processing, asset management, chat functionality, chat history, dashboard APIs, and search capabilities.

Frontend

In Progress

Current progress:

- Next.js project initialized
- Tailwind CSS configured
- Axios integrated
- Login page created

Upcoming work:

- Dashboard interface
 - File upload page
 - Asset listing page
 - AI Chat interface
 - Search page
 - User profile page
 - Responsive UI improvements
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Future Enhancements

- Multiple document formats (Word, Excel, Images)
 - OCR support for scanned PDFs
 - Voice-based document interaction
 - AI-powered semantic search
 - Document sharing between users
 - Role-based access control
 - Cloud storage integration
 - Email notifications
 - Document version control
 - Analytics dashboard
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Learning Outcomes

This project demonstrates practical implementation of:

- REST API development using FastAPI
- Authentication using JWT
- Database management with PostgreSQL
- AI integration using Google Gemini
- PDF processing and text extraction

- Full-stack application development
 - Frontend development using Next.js and Tailwind CSS
 - API integration with Axios
 - Secure document management
 - Modern software architecture and best practices
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Conclusion

AI Asset Vault is a modern intelligent document management system that combines secure asset storage with artificial intelligence to improve document accessibility and usability. Through features such as automated summaries, AI-powered question answering, intelligent tagging, and efficient search, the application offers a smarter alternative to conventional file management systems. Once the frontend and deployment phases are completed, the project will serve as a comprehensive full-stack AI application suitable for portfolio presentation and real-world use.